MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSCIAM200 - Principles of Robotics and Automation

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Robotics is mainly concerned with: A. Study of electrical circuits B.Design and application of robots C. Human resource management D. Computer networking	U	1
2	What does PLC stand for? A. Programmable Logic Circuit B. Programmable Line Controller C. Programmable Logic Controller D. Process Logic Controller	K	2
3	Which PLC type allows easy expansion by adding modules? A. Modular PLC B. Rack PLC C. Compact PLC D. Distributed PLC	U	2
4	Which device is connected to the PLC input module? A. Motor B. Lamp C. Relay coil D. Push button	U	2
5	PLC scans the program in which order? A Bottom to top B. Random order C. Top to bottom D. Right to left	U	2
6	Which programming language is most commonly used in PLC? A.C language B. Python C. Ladder Logic D. Java	K	2
7	Which of the following is an example of an open-loop control system? A. Automatic washing machine B. Traffic light controller C. Room temperature control D. Speed control of a car	U	3

8	Which element compares the output with the reference input? A. Actuator B. Comparator C. Sensor D. Controller	K	3
9	Which sensor is commonly used to detect the presence of objects on a conveyor belt? A. Temperature sensor B. Proximity sensor C. Pressure sensor D. Humidity sensor	U	3
10	The primary role of a control system in robotics is to: A. Design robot structure B. Control robot motion and actions C. Supply power to robot D. Store data	U	3

1 * 10 = 10

Part B

Short Answer Questions
Answer any 4 Questions
Each question carries 5 marks

11	Explain the working principle of a PIR sensor.	U	1
12	Define industrial automation and explain its purpose.	U	1
13	Describe the steps involved in interfacing a motor with PLC.	A	2
14	Explain the basic architecture of a PLC with neat explanation.	U	2
15	Explain the working principle of an automatic water level control system with a neat diagram.	U	3
16	Explain the role of industrial robots in spot welding operations.	U	3

4 * 5 = 20

Part C

Essay Questions
Answer any **2** Questions
Each question carries 10 marks

17	Explain stepper motor in detail.	U	1
18	Describe the working and applications of an IR proximity sensor.	U	1
19	Explain different types of PLCs with their advantages, and applications.	U	2
20	Describe how a pick-and-place robot can sort fruits into small, medium, and large categories.	A	3

2 * 10 = 20